Multi-Factor Equity Scoring and Portfolio Construction Engine

Production-Grade, Regime-Aware, Empirically-Weighted Extension

Internal Quantitative Research Specification

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1 Notation and Existing Core Engine (Unchanged)

We retain the original notation and all existing definitions without modification. [file:1]

1.1 Basic Notation

- i – stock index (single security). [file:1]
- t – time index (trading day). [file:1]
- s – sector index. [file:1]
- N – number of stocks in a given peer set (universe or sector). [file:1]
- $P_{i,t}$ – adjusted closing price of stock i on day t. [file:1]
- $R_{i,t}$ – simple daily return of stock i on day t:

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}}.$$

- W – lookback window length in trading days (e.g., $21 \approx$ 1M, $63 \approx$ 3M). [file:1]
- $\text{Ret}_{i,W}$ – cumulative simple return of stock i over the last W days. [file:1]
- μ_i, σ_i – annualized mean return and volatility of stock i. [file:1]
- R_f – annualized risk-free rate (e.g., 1Y sovereign yield). [file:1]
- S_s – peer set: all stocks in sector s. [file:1]
- $\text{Rank}(X_i)$ – ordinal rank of X_i within a peer set (1 = worst, N = best). [file:1]
- $\text{PctRank}(X_i)$ – percentile rank of X_i in $[0, 1]$. [file:1]
- $\text{WinsorizedRank}(X_i)$ – rank with tails clipped before percentile conversion. [file:1]
- $\text{Score}_i^{(k)}$ – generic notation for a factor score on $[0, 1]$. [file:1]
- $\bar{X}_i$ – time-series mean of metric X for stock i. [file:1]
- $\text{SD}(X_i)$ – time-series standard deviation of metric X for stock i. [file:1]
- MDD_i – maximum drawdown of stock i over a specified window. [file:1]

All scores used in composites (FundamentalScore, USPScore, MomentumScore, RiskAdjReturnScore, etc.) are normalized to $[0, 1]$ unless stated otherwise. [file:1]

1.2 Core Scoring and Data Construction (Unchanged)

The original engine defines:

- Point-in-time data construction for prices, fundamentals, sector codes, patents, and risk-free rate. [file:1]
- Winsorized-shrinkage percentile scoring backbone:

$$\text{PctRank}(X_i) = \frac{\text{Rank}(X_i) - 1}{N - 1}, \tag{1}$$

$$r_i^{(W)} = \min\{\max(r_i, r_L), r_U\}, \tag{2}$$

$$\text{PctRank}^{(W)}(X_i) = \frac{r_i^{(W)} - 1}{N - 1}, \tag{3}$$

$$\lambda_s = \frac{k}{k + n_s}, \quad k > 0, \tag{4}$$

$$\text{Score}_i^{(0)} = \text{PctRank}^{(W)}(X_i), \tag{5}$$

$$\text{Score}_i^{(k)} = (1 - \lambda_s)\text{Score}_i^{(0)} + \lambda_s \bar{\text{Score}}_s. \tag{6}$$

- Fundamental layer: ROE, growth (CAGR), margins, R&D, quarterly performance. [file:1]
- USP layer: strength, replaceability, durability, combined as:

$$\text{USPScore}_i = 0.4 \cdot \text{USPStrength}_i + 0.3 \cdot \text{ReplaceScore}_i + 0.3 \cdot \text{Durability}_i.$$

- Momentum layer with multi-horizon equal-weighted scores:

$$\text{MomentumScore}_i = \text{Momentum}_{1\text{D},i} + \text{Momentum}_{1\text{W},i} + \text{Momentum}_{1\text{M},i} + \text{Momentum}_{3\text{M},i} + \text{Momentum}_{6\text{M},i} + \text{M}$$

- Risk-adjusted performance layer: Sharpe, Sortino, Max Drawdown scores. [file:1]

1.3 Existing Composite Scores (Unchanged)

$$\text{FundamentalScore}_i = \frac{\text{ROEScore}_i + \text{GrowthScore}_i + \text{MarginScore}_i + \text{RnDScore}_i + \text{QuarterlyPerformanceScore}_i}{5}, \tag{7}$$

$$\text{StockScore}_i = 0.7 \cdot \text{FundamentalScore}_i + 0.3 \cdot \text{USPScore}_i, \tag{8}$$

$$\text{QualitativeScore}_i = 0.8 \cdot \text{StockScore}_i + 0.2 \cdot \text{MomentumScore}_i, \tag{9}$$

$$\text{DecisionScore}_i = 0.6 \cdot \text{QualitativeScore}_i + 0.4 \cdot \text{RiskAdjReturnScore}_i. \tag{10}$$

The empirical-weight determination section in the original text defines a t-statistics-based scheme; we retain it exactly and build on it in Section 2. [file:1]

2 Factor Weight Optimization and Optimized Weighted Score

We introduce an *additive* empirically-driven weighting layer on top of the existing equal-weight architecture, without modifying any of the original formulas. [file:1] The goal is to obtain a production-grade *OptimizedWeightedScore* based on ex-ante predictive efficacy of each factor group and sub-factor.

2.1 Factor Space and Notation

Let the factor index k run over all atomic factor scores entering the composites, for instance:

- $k \in \mathcal{F}$: fundamental sub-factors (ROE, CAGR, margins, R&D, quarterly metrics).
- $k \in \mathcal{U}$: USP sub-factors (USPStrength, ReplaceScore, Durability).
- $k \in \mathcal{M}$: momentum horizons (1D, 1W, 1M, 3M, 6M, 1Y).
- $k \in \mathcal{R}$: risk-adjusted measures (SharpeScore, SortinoScore, MaxDDScore).

For each stock i and rebalancing date T, we denote the corresponding normalized factor score by

$$\mathrm{Score}_{i,T}^{(k)} \in [0, 1].$$

We also denote forward excess returns over horizon H (e.g., one month) as

$$r_{i,T+H}^{\mathrm{fwd}} = R_{i,T+1:T+H} - R_{T+1:T+H}^{\mathrm{bench}},$$

where R^{bench} is the benchmark cumulative return. [file:1]

2.2 Predictive Efficacy Measures

For each factor k and evaluation date T, we define three predictive metrics over a rolling window of length L months (e.g., $L = 12$):

Information Coefficient (IC). For each month $u \in \{T - L + 1, \ldots, T\}$, compute the cross-sectional Pearson correlation between factor k and forward returns:

$$\mathrm{IC}_u^{(k)} = \mathrm{corr}_i\Big(\mathrm{Score}_{i,u}^{(k)},\, r_{i,u+H}^{\mathrm{fwd}}\Big). \tag{11}$$

Define the rolling average IC and volatility:

$$\overline{\mathrm{IC}}_T^{(k)} = \frac{1}{L} \sum_{u=T-L+1}^{T} \mathrm{IC}_u^{(k)}, \tag{12}$$

$$\sigma_{\mathrm{IC}}^{(k)} = \sqrt{\frac{1}{L-1} \sum_{u=T-L+1}^{T} \Big(\mathrm{IC}_u^{(k)} - \overline{\mathrm{IC}}_T^{(k)}\Big)^2}. \tag{13}$$

Rank Information Coefficient (Rank IC). Analogously, define cross-sectional Spearman rank correlations:

$$\mathrm{RankIC}_u^{(k)} = \mathrm{corr}_i\Big(\mathrm{rank}(\mathrm{Score}_{i,u}^{(k)}),\, \mathrm{rank}(r_{i,u+H}^{\mathrm{fwd}})\Big), \tag{14}$$

$$\overline{\mathrm{RankIC}}_T^{(k)} = \frac{1}{L} \sum_{u=T-L+1}^{T} \mathrm{RankIC}_u^{(k)}, \tag{15}$$

$$\sigma_{\mathrm{RankIC}}^{(k)} = \sqrt{\frac{1}{L-1} \sum_{u=T-L+1}^{T} \Big(\mathrm{RankIC}_u^{(k)} - \overline{\mathrm{RankIC}}_T^{(k)}\Big)^2}. \tag{16}$$

Predictive Correlation with Forward Returns. We also compute a simple absolute correlation of the factor with forward returns:

$$\rho_T^{(k)} = \left| \mathrm{corr}_{i,u \in [T-L+1,T]} \left(\mathrm{Score}_{i,u}^{(k)},\, r_{i,u+H}^{\mathrm{fwd}} \right) \right|. \tag{17}$$

2.3 Stability-Adjusted Predictive Strength

To capture both magnitude and stability of predictive power, we define a *predictive strength* score for each factor:

$$\Phi_T^{(k)} = \alpha_1 \cdot \frac{|\overline{\mathrm{IC}}_T^{(k)}|}{\sigma_{\mathrm{IC}}^{(k)} + \epsilon} + \alpha_2 \cdot \frac{|\overline{\mathrm{RankIC}}_T^{(k)}|}{\sigma_{\mathrm{RankIC}}^{(k)} + \epsilon} + \alpha_3 \cdot \rho_T^{(k)}, \tag{18}$$

where $\epsilon > 0$ is a small regularization constant and $(\alpha_1, \alpha_2, \alpha_3)$ are fixed hyperparameters reflecting the weighting philosophy (e.g., equal emphasis on IC and RankIC, with α_3 moderating raw correlation). [file:1]

By construction, $\Phi_T^{(k)} \geq 0$, and higher values indicate factors with both strong and stable predictive relationships to forward returns.

2.4 Factor Weights and Optimized Weighted Score

We define a normalized factor weight for each k:

$$\tilde{w}_k(T) = \max\{\Phi_T^{(k)}, 0\}, \tag{19}$$

$$w_k(T) = \frac{\tilde{w}_k(T)}{\sum_\ell \tilde{w}_\ell(T)}. \tag{20}$$

These $w_k(T)$ are re-estimated on a rolling basis (e.g., monthly), using only information available up to date T to preserve point-in-time integrity. [file:1]

The *optimized weighted factor score* for stock i is then defined as:

$$\mathrm{OptimizedWeightedScore}_{i,T} = \sum_k w_k(T) \cdot \mathrm{Score}_{i,T}^{(k)}. \tag{21}$$

Note that:

- All original composite scores (FundamentalScore, USPScore, MomentumScore, RiskAdjReturnScore, StockScore, QualitativeScore, DecisionScore) remain defined exactly as before and can be used in parallel. [file:1]
- $\mathrm{OptimizedWeightedScore}_{i,T}$ provides an additional, empirically-informed composite that can feed into regime-aware and portfolio construction layers (Sections 6 and 7) without invalidating the equal-weight baseline.

2.5 Illustrative Factor-Weight Table

For implementation and auditability, we recommend maintaining a time-stamped table of factor weights. An example LaTeX representation (for a single date T) is:

Table 1: Illustrative Factor Weights at Date T (Schematic)

Factor k	Group	Predictive Strength $\Phi_T^{(k)}$	Weight $w_k(T)$
ROEScore	Fundamental	$\Phi_T^{(\mathrm{ROE})}$	$w_{\mathrm{ROE}}(T)$
GrowthScore	Fundamental	$\Phi_T^{(\mathrm{Growth})}$	$w_{\mathrm{Growth}}(T)$
MarginScore	Fundamental	$\Phi_T^{(\mathrm{Margin})}$	$w_{\mathrm{Margin}}(T)$
RnDScore	Fundamental	$\Phi_T^{(\mathrm{RnD})}$	$w_{\mathrm{RnD}}(T)$
QuarterlyPerformanceScore	Fundamental	$\Phi_T^{(\mathrm{Qtr})}$	$w_{\mathrm{Qtr}}(T)$
USPStrength	USP	$\Phi_T^{(\mathrm{USPStr})}$	$w_{\mathrm{USPStr}}(T)$
ReplaceScore	USP	$\Phi_T^{(\mathrm{Replace})}$	$w_{\mathrm{Replace}}(T)$
Durability	USP	$\Phi_T^{(\mathrm{Dur})}$	$w_{\mathrm{Dur}}(T)$
$\mathrm{Momentum}_{1\mathrm{M}}$	Momentum	$\Phi_T^{(1\mathrm{M})}$	$w_{1\mathrm{M}}(T)$
$\mathrm{Momentum}_{3\mathrm{M}}$	Momentum	$\Phi_T^{(3\mathrm{M})}$	$w_{3\mathrm{M}}(T)$
$\mathrm{Momentum}_{6\mathrm{M}}$	Momentum	$\Phi_T^{(6\mathrm{M})}$	$w_{6\mathrm{M}}(T)$
$\mathrm{Momentum}_{12\mathrm{M}}$	Momentum	$\Phi_T^{(12\mathrm{M})}$	$w_{12\mathrm{M}}(T)$
SharpeScore	Risk-Adj	$\Phi_T^{(\mathrm{Sharpe})}$	$w_{\mathrm{Sharpe}}(T)$
SortinoScore	Risk-Adj	$\Phi_T^{(\mathrm{Sortino})}$	$w_{\mathrm{Sortino}}(T)$
MaxDDScore	Risk-Adj	$\Phi_T^{(\mathrm{MaxDD})}$	$w_{\mathrm{MaxDD}}(T)$

3 Momentum Reconstruction and Medium-Term Emphasis

We now introduce a refined momentum composite $\mathrm{MomentumScore}_i^*$ which coexists with the original $\mathrm{MomentumScore}_i$ in (1.2). [file:1] The original equal-weight formulation is retained unchanged and can continue to be used.

3.1 Existing Multi-Horizon Momentum (Unchanged)

For each horizon $W \in \{1\mathrm{D}, 1\mathrm{W}, 1\mathrm{M}, 3\mathrm{M}, 6\mathrm{M}, 1\mathrm{Y}\}$:

$$\mathrm{Ret}_{i,W} = \prod_{k=1}^{W}(1 + R_{i,t-k}) - 1, \tag{22}$$

$$\mathrm{MomentumWScore}_{i,W} = \mathrm{Score}(\mathrm{Ret}_{i,W} \text{ within } S_s \text{ or full universe}). \tag{23}$$

Then

$$\mathrm{MomentumScore}_i = \sum_W \mathrm{MomentumWScore}_{i,W}.$$

3.2 Refined Weighted Momentum Composite

Define a *weighted* momentum score focusing on medium-term horizons:

$$\mathrm{MomentumScore}_i^* = w_{1\mathrm{M}} \cdot M_{i,1\mathrm{M}} + w_{3\mathrm{M}} \cdot M_{i,3\mathrm{M}} + w_{6\mathrm{M}} \cdot M_{i,6\mathrm{M}} + w_{12\mathrm{M}} \cdot M_{i,12\mathrm{M}}, \tag{24}$$

where $M_{i,W} = \mathrm{MomentumWScore}_{i,W}$ are the existing normalized scores for the respective horizons. [file:1]

We explicitly constrain short-term weights:

$$w_{1\mathrm{D}} \approx 0, \tag{25}$$

$$w_{1\mathrm{W}} \approx 0, \tag{26}$$

so that intra-day and very short-term noise has negligible impact on the composite. [file:1] The medium-term emphasis is achieved by imposing:

$$w_{1\mathrm{M}}, w_{3\mathrm{M}}, w_{6\mathrm{M}}, w_{12\mathrm{M}} \geq 0, \tag{27}$$

$$w_{1\mathrm{M}} + w_{3\mathrm{M}} + w_{6\mathrm{M}} + w_{12\mathrm{M}} = 1. \tag{28}$$

3.3 Empirical Determination of Momentum Horizon Weights

In alignment with the factor-weight philosophy of Section 2, we can derive w_W via predictive strength of each horizon:

$$\Phi_T^{(\mathrm{Mom},W)} = \alpha_1 \frac{|\overline{\mathrm{IC}}_T^{(\mathrm{Mom},W)}|}{\sigma_{\mathrm{IC}}^{(\mathrm{Mom},W)} + \epsilon} + \alpha_2 \frac{|\overline{\mathrm{RankIC}}_T^{(\mathrm{Mom},W)}|}{\sigma_{\mathrm{RankIC}}^{(\mathrm{Mom},W)} + \epsilon} + \alpha_3 \rho_T^{(\mathrm{Mom},W)}, \tag{29}$$

$$\tilde{w}_W(T) = \Phi_T^{(\mathrm{Mom},W)}, \quad W \in \{1\mathrm{M}, 3\mathrm{M}, 6\mathrm{M}, 12\mathrm{M}\}, \tag{30}$$

$$w_W(T) = \frac{\tilde{w}_W(T)}{\sum_{U \in \{1\mathrm{M},3\mathrm{M},6\mathrm{M},12\mathrm{M}\}} \tilde{w}_U(T)}. \tag{31}$$

3.4 Momentum Weight Table

For transparency, an illustrative table:

Table 2: Illustrative Momentum Horizon Weights

Horizon W	Predictive Strength $\Phi_T^{(\mathrm{Mom},W)}$	Weight $w_W(T)$
1M	$\Phi_T^{(\mathrm{Mom},1\mathrm{M})}$	$w_{1\mathrm{M}}(T)$
3M	$\Phi_T^{(\mathrm{Mom},3\mathrm{M})}$	$w_{3\mathrm{M}}(T)$
6M	$\Phi_T^{(\mathrm{Mom},6\mathrm{M})}$	$w_{6\mathrm{M}}(T)$
12M	$\Phi_T^{(\mathrm{Mom},12\mathrm{M})}$	$w_{12\mathrm{M}}(T)$

4 Valuation Layer and ValueScore

We add a valuation layer that is fully consistent with the existing normalization backbone (winsorized percentile with empirical-Bayes shrinkage) and sector-relative comparisons. [file:1] This layer is additive and does not alter existing factor definitions.

4.1 Raw Valuation Metrics

For each stock i at time T:

- Price-to-Earnings ratio:

$$\mathrm{PE}_i = \frac{P_{i,T}}{\mathrm{EPS}_{i,T}},$$

where $\mathrm{EPS}_{i,T}$ is trailing or forward EPS as per a consistent, point-in-time-constrained data source.

- Enterprise Value to EBITDA:

$$\mathrm{EVEBITDA}_i = \frac{\mathrm{EV}_{i,T}}{\mathrm{EBITDA}_{i,T}}.$$

- Free Cash Flow Yield:

$$\mathrm{FCFYield}_i = \frac{\mathrm{FCF}_{i,T}}{\mathrm{MarketCap}_{i,T}}.$$

4.2 Normalization and Inverse Scoring

Each raw metric is converted to a sector-relative factor score using the same winsorized-shrinkage backbone in Section 2. [file:1]

PEScore (inverse multiple). Lower P/E is better (value-style), so we define:

$$\mathrm{PEScore}_i = \mathrm{Score}\left(-\mathrm{PE}_i \text{ within } S_s\right). \tag{32}$$

EV/EBITDA Score (inverse multiple). Similarly, lower EV/EBITDA is better:

$$\mathrm{EVEBITDA}_i^{\mathrm{Score}} = \mathrm{Score}\left(-\mathrm{EVEBITDA}_i \text{ within } S_s\right). \tag{33}$$

FCF Yield Score (direct yield). Higher FCF yield is better:

$$\mathrm{FCFYieldScore}_i = \mathrm{Score}\left(\mathrm{FCFYield}_i \text{ within } S_s\right). \tag{34}$$

All scores above are sector-relative, winsorized, and shrunk as per (1)–(6). [file:1]

4.3 Composite ValueScore

We define the valuation composite as:

$$\mathrm{ValueScore}_i = \frac{\mathrm{PEScore}_i + \mathrm{EVEBITDA}_i^{\mathrm{Score}} + \mathrm{FCFYieldScore}_i}{3}. \tag{35}$$

4.4 Valuation Weight Table

An example tabular representation of per-metric contributions:

This $\mathrm{ValueScore}_i$ can be incorporated into the regime-aware final score in Section 6.4, while the original DecisionScore remains unchanged and available. [file:1]

5 USP Layer Refinement with Conditional Weighting

We now refine the USP layer by introducing a conditional, sector-dependent weighting overlay on top of the existing fixed 0.4/0.3/0.3 combination in (1.2). [file:1] The original formula is preserved exactly; we define an additional $\mathrm{USPScore}_i^*$.

Table 3: Valuation Sub-Factors and Equal Weights

Metric	Score	Weight in ValueScore
Price-to-Earnings (inverse)	PEScore_i	$\frac{1}{3}$
EV/EBITDA (inverse)	$\text{EVEBITDA}_i^{\text{Score}}$	$\frac{1}{3}$
FCF Yield (direct)	FCFYieldScore_i	$\frac{1}{3}$

5.1 Existing USP Components (Unchanged)

Recall:

$$\text{USPStrength}_i \text{ (ROIC spread, margin level/stability, market share). [file:1]} \tag{36}$$

$$\text{ReplaceScore}_i \text{ (revenue stability, recurring revenue). [file:1]} \tag{37}$$

$$\text{Durability}_i \text{ (ROIC trend, long-term margin stability, patents). [file:1]} \tag{38}$$

$$\text{USPScore}_i = 0.4 \cdot \text{USPStrength}_i + 0.3 \cdot \text{ReplaceScore}_i + 0.3 \cdot \text{Durability}_i. [file:1] \tag{39}$$

5.2 Sector-Dependent Conditional Weighting

We define a refined sector-conditional USP score:

$$\text{USPScore}_i^* = w_1^{(s)} \cdot \text{USPStrength}_i + w_2^{(s)} \cdot \text{ReplaceScore}_i + w_3^{(s)} \cdot \text{Durability}_i, \tag{40}$$

with s the sector of stock i, such that:

$$w_1^{(s)}, w_2^{(s)}, w_3^{(s)} \geq 0, \tag{41}$$

$$w_1^{(s)} + w_2^{(s)} + w_3^{(s)} = 1. \tag{42}$$

Patent Weight in Non-Innovation Sectors. To reflect the economic irrelevance of patent intensity in certain sectors, we impose:

$$\text{PatentsScore}_i \to 0 \quad \text{(or treated as neutral 0.5) for non-innovation sectors,} \tag{43}$$

either by setting the patents component to a fixed neutral value in Durability_i or by assigning a vanishing weight to its contribution at the sector configuration stage. [file:1] In practice, this is implemented as a sector-specific Durability decomposition:

$$\text{Durability}_i^{(s)} = \begin{cases} \frac{\text{DurROICScore}_i + \text{DurMarginScore}_i + \text{PatentsScore}_i}{3}, & s \in \mathcal{S}_{\text{innovation}}, \\ \frac{\text{DurROICScore}_i + \text{DurMarginScore}_i}{2}, & s \notin \mathcal{S}_{\text{innovation}}. \end{cases}$$

5.3 Sector-Dependent USP Weight Table

We maintain a sector-by-component weight matrix, e.g.:

The baseline 0.4/0.3/0.3 combination remains available as USPScore_i, while USPScore_i^* offers a sector-tuned alternative. [file:1]

6 Regime-Aware Extension

We now formalize a regime-aware final scoring structure that extends the original DecisionScore with macro and market state dependence, building coherently on the macro overlay concepts in the existing MacroOverrideLayer. [file:1]

Table 4: Illustrative Sector-Dependent USP Weights

Sector s	$w_1^{(s)}$ (Strength)	$w_2^{(s)}$ (Replaceability)	$w_3^{(s)}$ (Durability)
Technology / Pharma	$w_1^{(\text{Tech})}$	$w_2^{(\text{Tech})}$	$w_3^{(\text{Tech})}$
Banks / NBFCs	$w_1^{(\text{Bank})}$	$w_2^{(\text{Bank})}$	$w_3^{(\text{Bank})}$
Retail / FMCG	$w_1^{(\text{Retail})}$	$w_2^{(\text{Retail})}$	$w_3^{(\text{Retail})}$
Utilities	$w_1^{(\text{Util})}$	$w_2^{(\text{Util})}$	$w_3^{(\text{Util})}$

6.1 Market Regime Definition

We define a discrete regime index $r \in \mathcal{R}$ based on three key market state variables:

Volatility. Let σ_{market} denote the realized volatility of a broad market index over a window W_σ (e.g., 30 days):

$$\sigma_{\text{market}} = \sqrt{252} \cdot \text{SD}\Big(R^{\text{index}}_{t-W_\sigma+1:t}\Big). \tag{44}$$

Trend. We use moving averages of the index price:

$$\text{MA}_{50} = \frac{1}{50}\sum_{k=1}^{50} P^{\text{index}}_{t-k}, \tag{45}$$

$$\text{MA}_{200} = \frac{1}{200}\sum_{k=1}^{200} P^{\text{index}}_{t-k}. \tag{46}$$

Drawdown. Let DD_{index} be the realized drawdown over a window W_{DD}:

$$\text{DD}_{\text{index}} = \max_{u\in[t-W_{\text{DD}},t]} \left(\frac{\max_{v\le u} P^{\text{index}}_v - P^{\text{index}}_u}{\max_{v\le u} P^{\text{index}}_v}\right). \tag{47}$$

6.2 Regime Classification Rules

We define regime r via rule-based thresholds on $(\sigma_{\text{market}}, \text{MA}_{50} - \text{MA}_{200}, \text{DD}_{\text{index}})$.

For example, a four-regime specification:

$$r = \begin{cases} \text{LowVol-UpTrend}, & \sigma_{\text{market}} \le \theta_\sigma^{\text{low}},\ \text{MA}_{50} > \text{MA}_{200},\ \text{DD}_{\text{index}} \le \theta_{\text{DD}}^{\text{mild}}, \\ \text{LowVol-DownTrend}, & \sigma_{\text{market}} \le \theta_\sigma^{\text{low}},\ \text{MA}_{50} \le \text{MA}_{200},\ \text{DD}_{\text{index}} \le \theta_{\text{DD}}^{\text{mild}}, \\ \text{HighVol-Correction}, & \sigma_{\text{market}} > \theta_\sigma^{\text{high}},\ \text{DD}_{\text{index}} \in (\theta_{\text{DD}}^{\text{mild}}, \theta_{\text{DD}}^{\text{severe}}], \\ \text{Crisis}, & \sigma_{\text{market}} > \theta_\sigma^{\text{high}},\ \text{DD}_{\text{index}} > \theta_{\text{DD}}^{\text{severe}}. \end{cases}$$

The thresholds $(\theta_\sigma^{\text{low}}, \theta_\sigma^{\text{high}}, \theta_{\text{DD}}^{\text{mild}}, \theta_{\text{DD}}^{\text{severe}})$ are design parameters to be calibrated by regime-segmented backtests, in line with the MacroOverrideLayer philosophy. [file:1]

6.3 Regime-Specific Factor Group Weights

We define regime-specific weights for each major group:

- F_i – fundamental quality proxy (we use $\text{FundamentalScore}_i$).
- M_i – momentum proxy (we use MomentumScore_i or MomentumScore_i^*).
- R_i – risk-adjusted performance proxy ($\text{RiskAdjReturnScore}_i$).
- V_i – valuation proxy (ValueScore_i).
- U_i – moat/USP proxy (USPScore_i or USPScore_i^*).

For each regime r, define:

$$w_F^{(r)},\ w_M^{(r)},\ w_R^{(r)},\ w_V^{(r)},\ w_U^{(r)} \geq 0, \tag{48}$$

$$w_F^{(r)} + w_M^{(r)} + w_R^{(r)} + w_V^{(r)} + w_U^{(r)} = 1. \tag{49}$$

6.4 Regime-Aware Final Score

For stock i in regime r, we define:

$$\text{FinalScore}_i^{(r)} = w_F^{(r)} F_i + w_M^{(r)} M_i + w_R^{(r)} R_i + w_V^{(r)} V_i + w_U^{(r)} U_i. \tag{50}$$

This $\text{FinalScore}_i^{(r)}$ coexists with the original DecisionScore_i in (10). The original score remains a regime-agnostic baseline, while (50) offers an explicit regime-aware layer consistent with the existing MacroOverrideLayer. [file:1]

6.5 Regime Weight Table

An example regime-by-weight table:

Table 5: Illustrative Regime-Specific Group Weights

Regime r	$w_F^{(r)}$	$w_M^{(r)}$	$w_R^{(r)}$	$w_V^{(r)}$	$w_U^{(r)}$
LowVol-UpTrend	$w_F^{(\text{LU})}$	$w_M^{(\text{LU})}$	$w_R^{(\text{LU})}$	$w_V^{(\text{LU})}$	$w_U^{(\text{LU})}$
LowVol-DownTrend	$w_F^{(\text{LD})}$	$w_M^{(\text{LD})}$	$w_R^{(\text{LD})}$	$w_V^{(\text{LD})}$	$w_U^{(\text{LD})}$
HighVol-Correction	$w_F^{(\text{HC})}$	$w_M^{(\text{HC})}$	$w_R^{(\text{HC})}$	$w_V^{(\text{HC})}$	$w_U^{(\text{HC})}$
Crisis	$w_F^{(\text{CR})}$	$w_M^{(\text{CR})}$	$w_R^{(\text{CR})}$	$w_V^{(\text{CR})}$	$w_U^{(\text{CR})}$

In practice, these weights can be chosen in accordance with the qualitative guidance already present in the MacroOverrideLayer (e.g., upweighting quality and risk-adjusted factors in rate-hike or high-volatility regimes). [file:1]

7 Portfolio Construction Layer

We now formalize the portfolio construction layer in a way that is fully compatible with the existing PortfolioContextLayer and diversification/sector-cap mechanisms. [file:1] The new layer uses $\text{FinalScore}_i^{(r)}$ as its core ranking variable but can be substituted with DecisionScore_i or $\text{OptimizedWeightedScore}_i$ if desired.

7.1 Selection Rule

Let $\text{FinalScore}_i^{(r)}$ denote the regime-specific final score for each stock i, as in (50). For a given date T and regime r, define the 90th percentile threshold $P_{90}^{(r)}$ of the cross-sectional distribution:

$$P_{90}^{(r)} = 90\text{th percentile of } \{\text{FinalScore}_j^{(r)}\}_j. \tag{51}$$

The selection set is:

$$\mathcal{I}_T^{(r)} = \left\{ i : \text{FinalScore}_i^{(r)} \geq P_{90}^{(r)} \right\}. \tag{52}$$

7.2 Naïve Proportional Position Sizing

Starting from unnormalized weights:

$$\tilde{w}_i^{(0)} = \text{FinalScore}_i^{(r)} \cdot \mathbf{1}\{i \in \mathcal{I}_T^{(r)}\}, \tag{53}$$

$$w_i^{(0)} = \frac{\tilde{w}_i^{(0)}}{\sum_{j \in \mathcal{I}_T^{(r)}} \tilde{w}_j^{(0)}}. \tag{54}$$

This realizes the design:

$$w_i = \frac{\text{FinalScore}_i}{\sum_j \text{FinalScore}_j}.$$

7.3 Portfolio Constraints

Per-Name Cap. Let $w_{\max}$ be a per-security cap (e.g., 5%):

$$w_i \leq w_{\max}.$$

Sector Caps. Let c_s be the maximum sector weight for sector s (e.g., $c_s = 25\%$ for core sectors, 15% for cyclicals, 20% for defensives as in the existing PortfolioContextLayer). [file:1] Let $\mathcal{I}_s$ denote the indices in sector s; then:

$$\sum_{i \in \mathcal{I}_s} w_i \leq c_s.$$

Feasible Weight Adjustment. We define the constrained optimization problem:

$$\max_{\{w_i\}} \quad \sum_{i \in \mathcal{I}_T^{(r)}} \text{FinalScore}_i^{(r)} \cdot w_i, \tag{55}$$

$$\text{s.t.} \quad \sum_{i \in \mathcal{I}_T^{(r)}} w_i = 1, \tag{56}$$

$$0 \leq w_i \leq w_{\max}, \quad \forall i \in \mathcal{I}_T^{(r)}, \tag{57}$$

$$\sum_{i \in \mathcal{I}_s \cap \mathcal{I}_T^{(r)}} w_i \leq c_s, \quad \forall s. \tag{58}$$

Since the objective is linear and the constraints are linear, this is a small linear program that can be solved efficiently daily. A simpler heuristic consistent with the existing PortfolioContextLayer is to apply multiplicative sector scaling:

$$w_i^{(\text{sector})} = w_i^{(0)} \cdot \min\left\{1,\ \frac{c_s}{\sum_{j \in \mathcal{I}_s \cap \mathcal{I}_T^{(r)}} w_j^{(0)}}\right\}, \quad i \in \mathcal{I}_s, \tag{59}$$

$$w_i = \frac{w_i^{(\text{sector})}}{\sum_{j \in \mathcal{I}_T^{(r)}} w_j^{(\text{sector})}}. \tag{60}$$

This mirrors the existing "AdjustedScore" sector cap logic while ensuring full investment. [file:1]

7.4 Transaction Costs and Net Returns

Let GrossReturn denote the portfolio gross return over a period and Turnover be the fraction of the portfolio traded (based on L^1 or L^+L^- norms of weight changes). [file:1] Let CostRate be the per-unit transaction cost.

We define:

$$\text{NetReturn} = \text{GrossReturn} - \text{CostRate} \cdot \text{Turnover}. \tag{61}$$

For a discrete rebalance from weights w_i^{old} to w_i^{new}:

$$\text{Turnover} = \frac{1}{2} \sum_i |w_i^{\text{new}} - w_i^{\text{old}}|. \tag{62}$$

7.5 Portfolio-Level Diversification and Consistency

The diversification metric defined in the original PortfolioContextLayer is reused unchanged:

$$\text{DiversificationScore} = 1 - \sum_i \sum_j w_i w_j \rho_{ij}, \tag{63}$$

where ρ_{ij} is the correlation matrix estimated over a recent window. [file:1] This provides a cross-check on the concentrated exposures implied by the $\text{FinalScore}_i^{(r)}$ selection and sizing rule.

8 Implementation-Ready Mathematical Flow

We now summarize an end-to-end, implementation-ready flow that integrates all layers while preserving the original architecture.

8.1 Daily Pipeline (Point-in-Time)

On each date T:

1. **Data Load (Point-in-Time).**
    - Load prices, fundamentals, sector codes, patents, macro signals, and risk-free rate with strict as-of dating. [file:1]

2. **Raw Metric Computation.**
    - Compute returns, volatility, drawdowns, correlations, ROE, CAGR, margins, R&D, quarterly KPIs, ROIC, WACC, patents, valuation metrics (PE, EV/EBITDA, FCFYield). [file:1]
3. **Normalization Backbone.**
    - For each metric X, compute $\text{Score}_i^{(k)}$ via winsorized percentile and empirical-Bayes shrinkage within sector S_s (Equations (1)–(6)). [file:1]
4. **Factor Layers.**
    - $\text{FundamentalScore}_i$ as in (7) (unchanged).
    - USPStrength, ReplaceScore, Durability, USPScore_i as in the original specification. [file:1]
    - MomentumScore_i per (3.1); additionally, compute MomentumScore_i^* per (24).
    - $\text{RiskAdjReturnScore}_i$ from SharpeScore, SortinoScore, MaxDDScore. [file:1]
    - ValueScore_i per (35).
5. **Composite Scores.**
    - StockScore_i, $\text{QualitativeScore}_i$, DecisionScore_i as in (8)–(10) (unchanged).
    - USPScore_i^* per (40) with sector-dependent weights.
    - $\text{OptimizedWeightedScore}_i$ per (21).
6. **Regime Detection.**
    - Compute σ_{market}, MA_{50}, MA_{200}, DD_{index} via (44)–(47), classify regime r using the rules in Section 6.
7. **Regime-Aware Final Score.**
    - For each stock i, compute F_i, M_i, R_i, V_i, U_i from the appropriate group proxies.
    - Compute $\text{FinalScore}_i^{(r)}$ via (50).
8. **Selection and Sizing.**
    - Determine $P_{90}^{(r)}$ and selection set $\mathcal{I}_T^{(r)}$ via (51)–(52).
    - Compute naive $w_i^{(0)}$ via (54), then apply per-name and sector caps via (7.3)–(60).
9. **Transaction Costs and Net Returns.**
    - Given old weights w_i^{old} and new w_i, compute Turnover via (62) and NetReturn via (61).
10. **Portfolio Context and Diagnostics.**
    - Compute DiversificationScore via (63), sector exposures, and monitor against policy thresholds. [file:1]
11. **Audit Trail.**
    - Persist all inputs, intermediate scores, final scores, regimes r, weights $w_k(T)$, $w_W(T)$, and portfolio weights w_i, as in the original roadmap. [file:1]

8.2 Backtest and Validation Integration

All newly introduced weights and regimes must be estimated using only historical data available at each evaluation date, following the same look-ahead-safe principles and validation protocols (walk-forward, regime-segmented analysis, turnover and cost modeling) already specified in the original document. [file:1]